

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

QIAN QI

ABSTRACT. The weighted cotangent dual is the vector space of invariant signed Radon measures, with probabilities as its positive normalized slice. Equality of the full perturbed pressure functional is treated platform by platform, never inferred from one scalar value. Spectral response and physical memory are placed on separately constructed common-domain bundles, and source perturbations, state tangents, optional projections and quadratic BSDE limits are rigorously typed.

Platform identifier: TYPED-FORM-BUNDLE-v5. **Controlling revision:** ROUND13-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** C2_INVARIANT_DUAL_COMMON_FORM_BUNDLE.tex.

CONTENTS

1. Round-thirteen positive closure: invariant duals and a common-domain response bundle	1
1.1. The invariant-integral quotient	2
1.2. Full pressure-functional rigidity	2
1.3. A common closed form domain	3
1.4. Resolved projections and memory response	3
1.5. Optional projections, Girsanov, and the entropic BSDE	4
1.6. Source and state chain rules	4
Scope and verification status	5

1. ROUND-THIRTEEN POSITIVE CLOSURE: INVARIANT DUALS AND A COMMON-DOMAIN RESPONSE BUNDLE

Invariant-integral equivalence, pressure equivalence, spectral response, and state contraction are four different notions. They are stated separately. The phase-dependent generators are compared through a common closed form

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 60F10, 60H10; Secondary 37A50, 49L25.

Key words and phrases. path-space large deviations, cotangent fields, entropic rigidity, tangent laws, Girsanov, BSDE.

domain built from the cylinder algebra; no finite correlation matrix is used to identify entire mutually singular L^2 spaces.

1.1. The invariant-integral quotient. Let $\mathcal{A}_W$ be the weighted dynamically Hölder cylinder algebra and let

$$\mathcal{N}_{\text{inv}} = \overline{\text{span}}\{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \in \mathbb{Q}_+\}^{\mathcal{A}_W}.$$

Constants are not included in $\mathcal{N}_{\text{inv}}$.

Theorem 1.1 (Signed invariant-Radon dual). *The continuous dual of $\mathcal{A}_W/\mathcal{N}_{\text{inv}}$ is the vector space of finite invariant signed Radon measures with finite W -moment. Invariant probabilities form its normalized positive slice. Consequently*

- (i) $\int F d\mu = \int G d\mu$ for every invariant probability iff $F - G \in \mathcal{N}_{\text{inv}}$;
- (ii) the two integrals differ by one common constant c for every invariant probability iff $F - G - c1 \in \mathcal{N}_{\text{inv}}$.

Proof. A finite signed invariant Radon measure annihilates every time coboundary and therefore acts on the quotient. Conversely, lift a quotient functional to $\mathcal{A}_W$. Weighted strict duality represents it by a countably additive signed Radon measure. Average its time translates. The uniform Lyapunov bound proved on each platform makes these averages weighted-tight; every limit is invariant and preserves the pairing. Hahn–Banach separation then shows that the annihilator of all invariant signed measures is exactly the closed coboundary space. Evaluating constants gives the two distinct statements (i)–(ii). $\square$

1.2. Full pressure-functional rigidity. For a potential F , write

$$\mathcal{P}_F(H) = P(F + H) - P(F)$$

after fixing the platform and one regular phase chart.

Theorem 1.2 (Platform-typed pressure rigidity). *Suppose $\mathcal{P}_F(H) = \mathcal{P}_G(H)$ for every H in a neighborhood of zero.*

- (i) *On the mixing Sinai symbolic/suspension chart, $F - G = c + U - U \circ \Theta$ in the weighted Hölder closure.*
- (ii) *On the finite-horizon hard-sphere source chart, the density/contact source difference is a constant plus the closed B3 endpoint–balance gauge, including the direct contact coordinate.*

Equality of one scalar pressure value is not used.

Proof. Differentiating the full perturbation identity at every nearby H shows that all nearby equilibrium linear functionals coincide. On the Sinai chart, periodic closing and the returned-branch spectral theorem imply equality of all periodic sums; the weighted Livšic theorem gives the constant plus coboundary representation. On the hard-sphere finite-time chart, strict convexity of the exact cumulant on the B3 quotient says that its full family of

first and second variations separates every class except the closed endpoint–balance gauge. The contact coefficient is $\Delta p + \psi$, so pure contact directions are included. This yields (ii) without invoking a hyperbolic Livšic theorem for hard spheres. $\square$

1.3. A common closed form domain. Let η range over a compact regular phase chart. Start from one common cylinder algebra $\mathcal{C}$. On $\mathcal{C}$ define

$$\|F\|_{\mathcal{V}}^2 = \sup_{\eta} \left(\|F\|_{L^2(\pi_{\eta})}^2 + \mathfrak{a}_{\eta}(F, F) + \|F\|_{\text{tr}, \eta}^2 \right),$$

where $\mathfrak{a}_{\eta}$ is the Doob/kinetic graph form and the null modes are removed before completion. Let $\mathcal{V}$ be the completion and let $\mathcal{V} \hookrightarrow \mathcal{H}_{\eta} \hookrightarrow \mathcal{V}^*$ be the resulting continuous field of Hilbert spaces.

Theorem 1.3 (Common-domain type-(B) bundle). *On the regular chart, the norms above are uniformly equivalent on $\mathcal{C}$, each form $\mathfrak{a}_{\eta}$ is closed sectorial on the common domain $\mathcal{V}$, and $\eta \mapsto \mathfrak{a}_{\eta}(F, G)$ is C^q for F, G in the corresponding fresh strong levels. The associated resolvents satisfy, as maps $\mathcal{V}^* \rightarrow \mathcal{V}$,*

$$D_{\eta}(z - L_{\eta})^{-1} = (z - L_{\eta})^{-1}(D_{\eta}L_{\eta})(z - L_{\eta})^{-1},$$

where $D_{\eta}L_{\eta}$ is interpreted as a bounded form $\mathcal{V} \rightarrow \mathcal{V}^*$.

Proof. A2/A4 give uniform graph and transfer estimates on the Sinai chart; B2/B3 give the corresponding transport/contact energy estimates on the hard-sphere chart. On a compact regular chart these bounds make the displayed norms equivalent and the cylinder algebra complete to one common Gelfand domain. Closedness and sectoriality follow by the uniform form estimate. Material source derivatives are bounded from the next strong level into $\mathcal{V}^*$, which gives type-(B) differentiability. Differentiate the weak resolvent equation

$$z\langle u, v \rangle + \mathfrak{a}_{\eta}(u, v) = \langle f, v \rangle$$

for every $v \in \mathcal{V}$ to obtain the typed formula. $\square$

1.4. Resolved projections and memory response. Choose physical observables $g_1, \dots, g_d \in \mathcal{C}$ after quotienting constants, conservation laws, and covariance null modes. Their correlation Gram matrix is uniformly positive on the chosen chart. Let R_{η} be the orthogonal projection onto their span in $\mathcal{H}_{\eta}$.

Theorem 1.4 (Covariant compressed-memory derivative). *The finite-rank sections R_{η} are C^q , and on the common form bundle*

$$C_{\eta}(z) = R_{\eta}(z - L_{\eta})^{-1}R_{\eta}, \quad \widehat{K}_{\eta}(z) = zR_{\eta} - R_{\eta}L_{\eta}R_{\eta} - C_{\eta}(z)^{-1}$$

have typed derivatives obtained by the product rule and Theorem 1.3. After adjoining the A4 descriptor modes, the regular memory kernel and all declared source derivatives decay exponentially.

Proof. Uniform positivity of the finite Gram matrix gives a smooth inverse and hence a smooth finite-rank projection section. The form-resolvent derivative is bounded $\mathcal{V}^* \rightarrow \mathcal{V}$; composition with the finite-rank sections is therefore an ordinary matrix-valued differentiable function. Differentiate C^{-1} by $D(C^{-1}) = -C^{-1}(DC)C^{-1}$. A4 gives integrable vertical bounds after all pole/zero Jordan chains are promoted, and these bounds are stable under the finite number of form derivatives. $\square$

1.5. Optional projections, Girsanov, and the entropic BSDE. Assume the A4 conditional kernels converge on a determining weighted cylinder class, the finite likelihood martingales are uniformly bounded in $L^{1+\delta}$, and they converge jointly with the driving martingale rough path. These hypotheses are now proved in A4 on the regular chart.

Theorem 1.5 (Filtration-stable likelihood limit). *The optional projections of the microscopic likelihoods onto the resolved history filtration converge stably to*

$$\mathcal{E} \left(\int_0^t \theta_s^T dW_s \right).$$

Under the tilted law the limiting resolved diffusion has the corresponding Girsanov drift. For bounded terminal value Φ , the pair (Y, Z) is the unique bounded solution of

$$Y_t = \Phi(X_T) + \int_t^T \frac{1}{2} |Z_s|^2 ds - \int_t^T Z_s dW_s,$$

after absorbing the chosen risk coefficient into Z and the driver.

Proof. Conditional-kernel convergence gives extended weak convergence of the resolved filtrations on the determining class. Joint martingale convergence and the $L^{1+\delta}$ bound pass conditional expectations and optional projections to the limit. The limiting exponential is a true martingale and Girsanov's theorem identifies the drift. Exponential transformation of the bounded terminal martingale gives the displayed quadratic BSDE; boundedness and the standard convex energy estimate give uniqueness in $\mathcal{S}^\infty \times \text{BMO}$. $\square$

1.6. Source and state chain rules. Let $\mathcal{C} : X \rightarrow Y$ be a typed continuous contraction. For a source $\varphi : Y \rightarrow \mathbb{R}$,

$$Q_{\mathcal{C}}(\varphi) = Q(\varphi \circ \mathcal{C}).$$

Proposition 1.6 (Separated contraction derivatives). *A source perturbation $h : Y \rightarrow \mathbb{R}$ satisfies*

$$DQ_{\mathcal{C}}(\varphi)[h] = DQ(\varphi \circ \mathcal{C})[h \circ \mathcal{C}].$$

If instead the state is varied through x_s with tangent $\dot{x}$, then

$$\frac{d}{ds} \varphi(\mathcal{C}x_s)|_{s=0} = D\varphi(\mathcal{C}x)D\mathcal{C}(x)\dot{x}.$$

These are different derivatives and are never combined into one formula.

Proof. The first identity is the Fréchet chain rule on the source Banach space; the second is the ordinary state-space chain rule. Their domains are distinct. $\square$

Theorem 1.7 (Round-thirteen C2 closure). *The series has a correctly typed invariant signed-measure cotangent dual, platform-specific full-pressure rigidity, a common-domain form bundle for source and memory response, a filtration-stable likelihood/Girsanov/BSDE limit, and separated source/state contraction derivatives.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.4, and 1.5, with Proposition 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`